

Operational Modal Identification of Hydraulic Gates under Varying Flow Conditions

شناسایی مودال عملیاتی دریاچه‌های هیدرولیکی در شرایط متغیر جریان

Candidate: **Zina** Proposed supervisor: **Prof. Reza Tarinejad** October 2026

Status

Preliminary concept prepared for discussion, subject to supervisor confirmation. Two points remain open: (1) a short recent-literature confirmation review of the gap; (2) confirmation that the scope does not overlap with current students' work. This concept is one half of a proposed paired programme and is independent of Jamal's concept.

The problem in plain terms

Operational modal analysis (OMA) estimates natural frequencies, mode shapes and damping from response alone. For a hydraulic gate these "apparent" properties are not fixed. When water level, submergence or opening changes, the hydrodynamic added mass and damping change the frequencies. At the same time, the exciting flow changes its own frequency content and may contain narrow-band vortex-shedding components.

The scientific problem is therefore: **how can the structural modal properties of a hydraulic gate be identified reliably when the surrounding water and the operating state themselves change the apparent dynamic properties?** Without an answer, a frequency shift measured on a gate cannot be attributed with confidence to damage, to a different water level, or to a flow-excitation peak.

Research gap

Prof. Tarinejad's recent work develops **transmissibility-based OMA**: weighted transmissibility [1, 2] and multi-reference weighted Hermitian transmissibility [4]. These methods find structural poles where transmissibility functions measured under **different loading conditions** coincide, which makes them less sensitive to the unknown input spectrum. That is attractive for coloured flow excitation. Recent work also notes that conventional OMA relies on white-noise, stationary-input assumptions [3].

These formulations assume that the **structure is the same** across the loading conditions used; the multi-reference approach [4] relaxes the requirements on the loading, not on the structure. In a hydraulic gate, the natural way to obtain different loading conditions (changing head, opening or submergence) also changes the structure's effective mass and damping, which violates the assumption at the point where the method relies on it. In a targeted check, no transmissibility-based or operating-state-parameterised OMA of hydraulic gates was found. This appears insufficiently addressed, subject to a structured confirmation review.

How the proposed approach differs from conventional OMA

Aspect	Conventional OMA (incl. current transmissibility methods)	Proposed approach
System assumption	Time-invariant structure	Modal properties depend on a measured hydraulic state θ (e.g. submergence ratio, head)
Input assumption	White, stationary (SSI/FDD); or several loadings on one unchanged structure (transmissibility)	Coloured flow excitation that changes with θ , including narrow-band components
Pole model	Fixed poles λ_r	Pole trajectories $\lambda_r(\theta)$ estimated across operating states
Spurious peaks	Removed by stabilisation or clustering	Separated physically: structural poles follow added mass; excitation peaks follow flow velocity

Core question and working hypothesis

Core scientific question

How can output-only identification separate genuine structural poles from operating-state-induced pole shifts and from flow-excitation peaks in a water-interacting structure?

Working hypothesis (to be tested)

When standard transmissibility OMA or SSI-COV pools data across operating states, it gives biased frequencies and spurious poles at flow-excitation peaks. If the poles are modelled as smooth functions $\lambda r(\theta)$ of a measured hydraulic state θ , the identified frequencies and damping should agree with input-output (EMA) ground truth across the operating range. The target accuracy, to be refined, is about 2% for frequency and about 20% for damping. Structural poles and excitation peaks should be distinguishable by how they scale.

Methodology

Work package	Content	Output
WP1 Theory	Extend the weighted/Hermitian transmissibility pole indicator [1, 4] to parameter-varying systems. Derive when transmissibility differences vanish at θ -dependent poles. Compare local (frozen- θ) and global (smooth-in- θ) estimators.	Formulation; Paper Z1
WP2 Numerical benchmark	FE model of a vertical-lift gate (plate and stiffeners) with submergence-dependent added mass. Driven by synthetic broadband plus narrow-band loads defined independently from the literature. Monte Carlo study of noise, record length and mode spacing.	Exact-pole benchmark; Paper Z1
WP3 Laboratory test-bed	One small flexible gate specimen in a flume. Still-water EMA (impact hammer) at 4–6 water levels. Operational runs under flow at 4–6 operating states.	Ground-truth validation; Paper Z2
WP4 Generality	Apply the method to public multi-event strong-motion records of an instrumented dam with different reservoir levels across events (e.g. Pacoima Dam, previously studied in [3]), with θ = reservoir level.	Field-scale demonstration; Paper Z3

Validation

- **Numerical:** identified poles compared with exact FE poles at each θ , under controlled noise.
- **Laboratory:** identified poles compared with EMA ground truth measured on the same specimen at the same water level.
- **Field-scale:** identified dam modes compared with published identifications. Benchmarks: standard transmissibility OMA, a multi-reference method [4] and SSI-COV [3].
- Validation does **not** use any result from Jamal's thesis.

Expected contribution and papers

If the hypothesis holds, the thesis would provide an identification method that treats the hydraulic operating state as an explicit parameter of the modal model. This would extend Prof. Tarinejad's transmissibility line from seismic records of dams to flow-excited hydraulic structures, together with a validated test protocol for gates. The novelty is subject to the confirmation review.

Paper	Indicative content	Candidate journals
Z1	Parameter-varying transmissibility formulation and numerical benchmark	Mech. Syst. Signal Process.; J. Sound Vib.
Z2	Laboratory validation on a flow-excited gate with EMA ground truth	J. Fluids Struct.; Struct. Control Health Monit.
Z3	Dam application with reservoir-level-dependent records	Earthq. Eng. Struct. Dyn.; J. Vib. Control

Independence and minimum requirements

- Uses its **own flexible specimen, accelerometers and EMA**, and **synthetic loads**. It does not need Jamal's pressure or load results.
- Year 1 (theory and FE) needs no laboratory, and Paper Z3 uses public records, so the thesis remains executable if Jamal's work is delayed.
- **Essential:** flume able to hold still water at several levels and run several flow states; one flexible gate specimen; 6–8 accelerometers; impact hammer with force sensor; synchronised DAQ (≥ 8 channels). Useful: strain gauges. Optional: laser vibrometer; field test on a regulator gate.

Title options

Discussion	Operational Modal Identification of Hydraulic Gates under Varying Flow Conditions
Formal PhD	Operating-State-Dependent Operational Modal Analysis of Flow-Excited Hydraulic Gates Using Transmissibility Functions
Publication	Parameter-varying transmissibility-based identification of hydraulic gates with water-level-dependent added mass

References (supervisor's publications)

- [1] Damadipour M., Tarinejad R., Aminfar M.H. (2021). Seismic modal identification using a new approach based on weighted transmissibility. *Earthquake Engineering & Structural Dynamics*, 50(7). <https://doi.org/10.1002/eqe.3435>
- [2] Damadipour M., Tarinejad R., Aminfar M.H. (2022). Structural system identification based on combining weighted transmissibility and wavelet transform. *Structural Control and Health Monitoring*, 29(2), e2868. <https://doi.org/10.1002/stc.2868>
- [3] Tarinejad R., Hafezi A., Pourgholi M. (2025). Automated modal analysis of an arch dam using a hybrid SSI framework based on error analysis and fuzzy clustering. *Journal of Vibration and Control*.
- [4] Tarinejad R., Amanzad F. (2025). Identification of operational modal parameters using multi-reference weighted Hermitian transmissibility functions. *Journal of Sound and Vibration*, 615, 119212. <https://doi.org/10.1016/j.jsv.2025.119212>